

Class 11: Structural Bioinformatics pt2

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AlphaFold Data Base (AFDB)

The EBI maintains the largest database of AlphaFold structure prediction models at:
<https://alphafold.ebi.ac.uk/>

From last class (before halloween) we saw that the PDB had 244,290 (Oct 2025).

The total number of protein sequence in UniProtKB is 199,579,901

Key point: This is a tiny fraction of sequence space that has structural coverage (0.12%).

244290/199579901 * 100

[1] 0.1224021

AFDB is attempting to address this gap...

There are two “Quality Scores” from AlphaFold one for residues (i.e. each amino acid) called **pLDDT** score. The other **PAE** score measures the confidence in the relative position of two residues (i.e. score for every pair of residues).

Generating your own structure predictions

Figure of 5 generated HIV-PR models

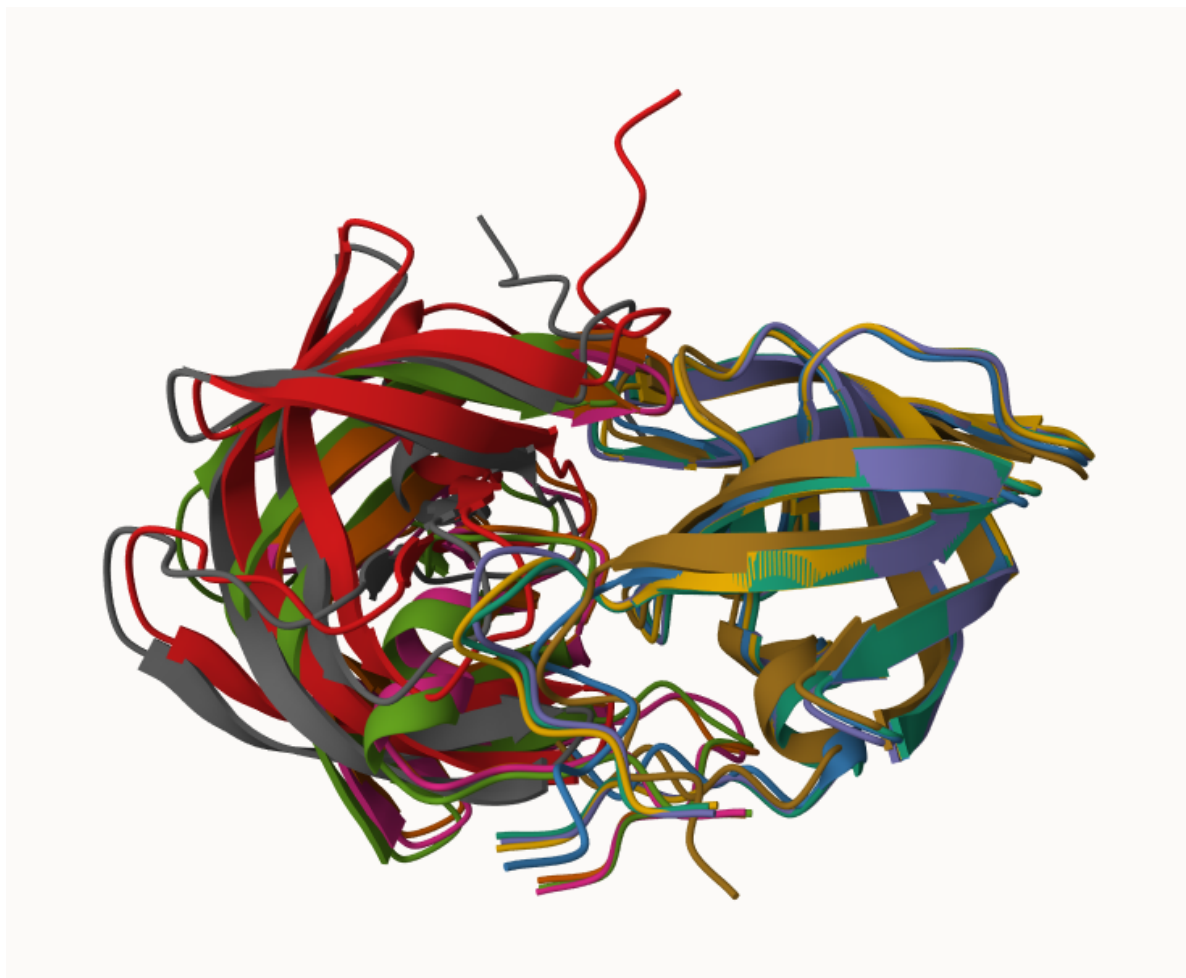


Figure of the top model

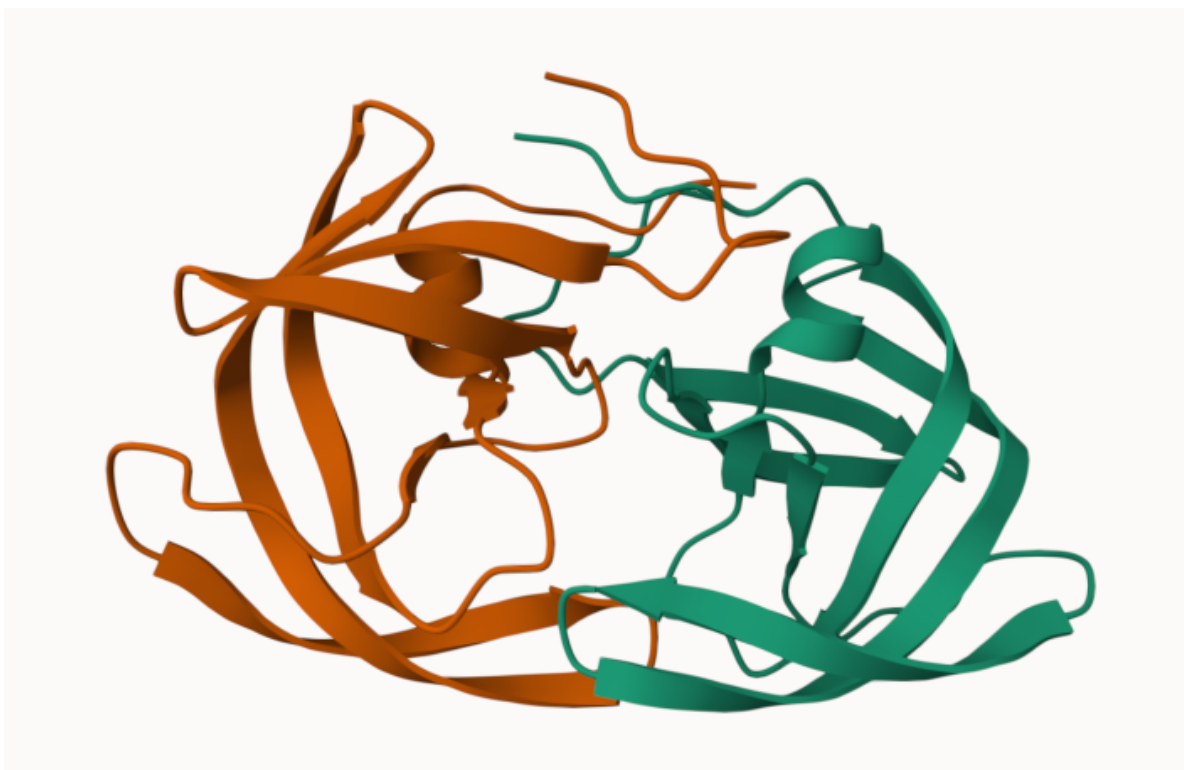
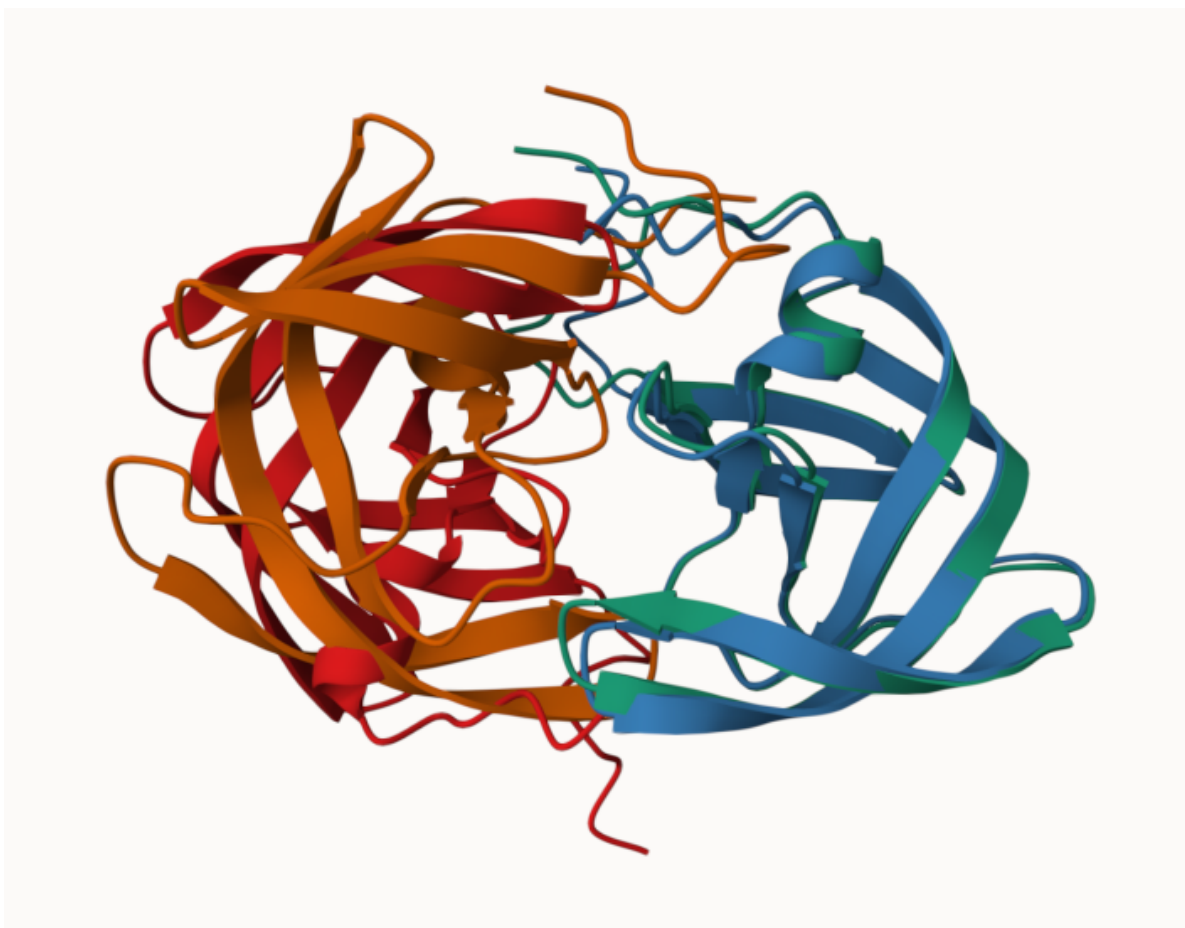
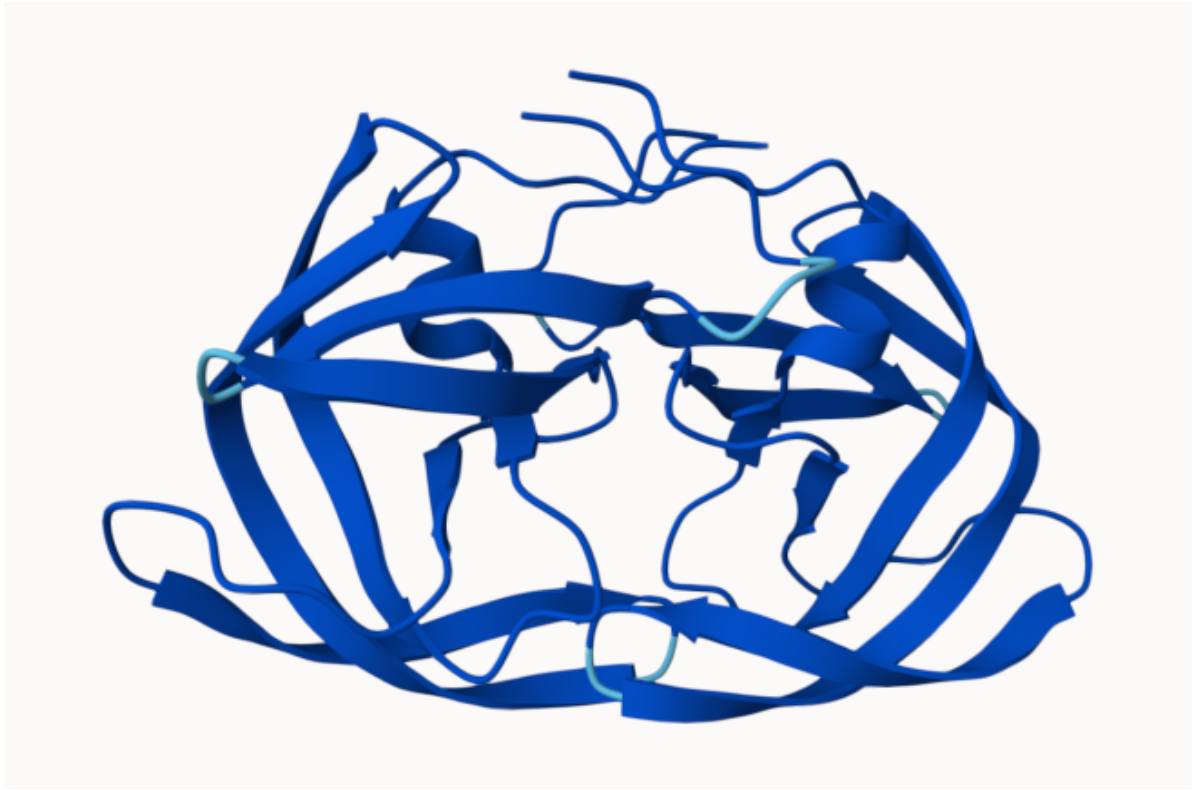


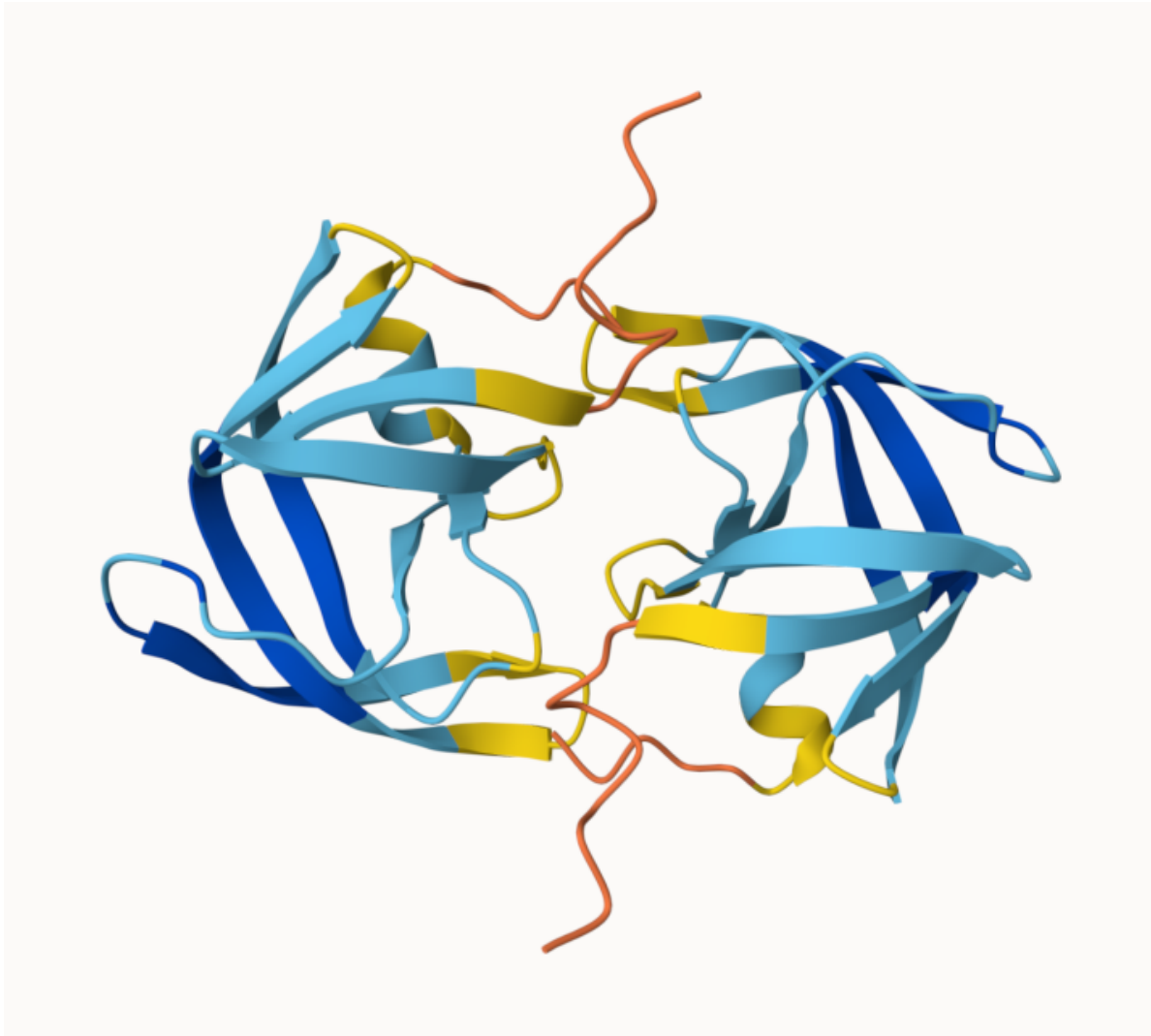
Figure of the top model and the last model



pLDDT score for top model



pLDDT score for the bottom model



Custom analysis of resulting models in R

Read key result files into R. The first thing I need to know is what my results directory/folder is called (i.e. its name is different for every AlphaFold run/job)

```
results_dir <- "HIVPR_dimer_23119/"
```

```
# File names for all PDB models
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
```

```
# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)
```

```
# Read all data from Models
# and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....
```

Extracting sequences

```
pdb/seq: 1    name: HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
pdb/seq: 2    name: HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
pdb/seq: 3    name: HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
pdb/seq: 4    name: HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
pdb/seq: 5    name: HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

RMSD is a standard measure of structural distance between coordinate sets. We can use the `rmsd()` function to calculate the RMSD between all pairs models.

```
rd <- rmsd(pdbs, fit=T)
```

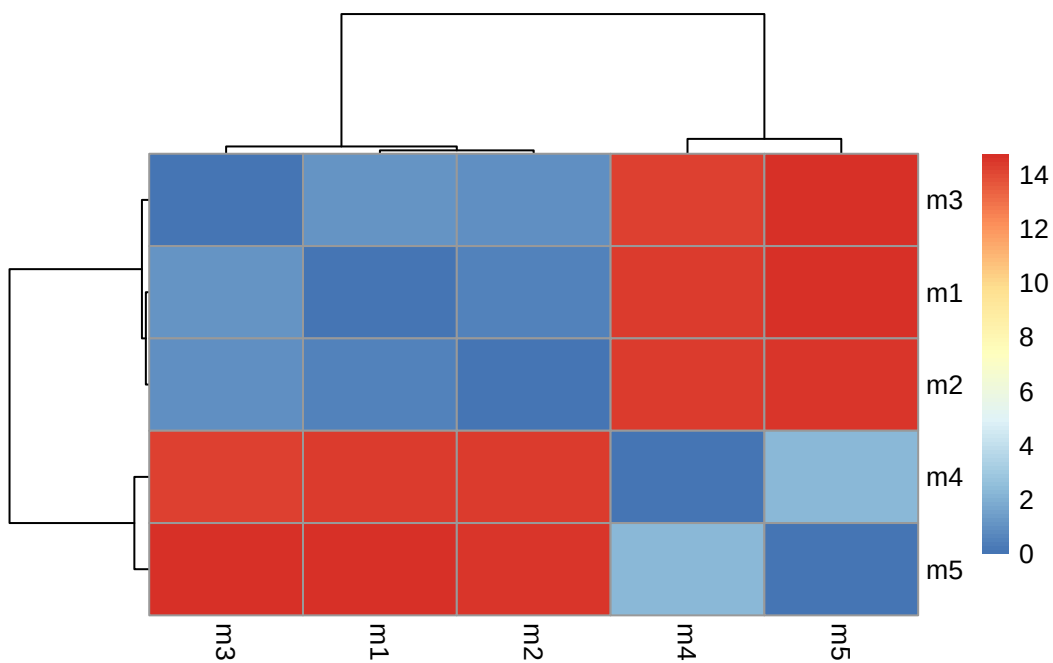
Warning in `rmsd(pdbs, fit = T)`: No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 14.754
```

Draw a heat map

```
library(pheatmap)  
  
colnames(rd) <- paste0("m",1:5)  
rownames(rd) <- paste0("m",1:5)  
pheatmap(rd)
```



```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

We can improve the superposition/fitting of our models by finding the most consistent “rigid core” common across all the models. For this we will use the `core.find()` function:


```
core <- core.find(pdb)
```

```
core size 197 of 198 vol = 9885.822
core size 196 of 198 vol = 6896.71
core size 195 of 198 vol = 1337.847
core size 194 of 198 vol = 1040.67
core size 193 of 198 vol = 951.857
core size 192 of 198 vol = 899.083
core size 191 of 198 vol = 834.732
core size 190 of 198 vol = 771.338
core size 189 of 198 vol = 733.065
core size 188 of 198 vol = 697.28
core size 187 of 198 vol = 659.742
core size 186 of 198 vol = 625.273
core size 185 of 198 vol = 589.541
core size 184 of 198 vol = 568.253
core size 183 of 198 vol = 545.015
core size 182 of 198 vol = 512.889
core size 181 of 198 vol = 490.723
core size 180 of 198 vol = 470.266
core size 179 of 198 vol = 450.731
core size 178 of 198 vol = 434.735
core size 177 of 198 vol = 420.337
core size 176 of 198 vol = 406.658
core size 175 of 198 vol = 393.334
core size 174 of 198 vol = 382.395
core size 173 of 198 vol = 372.858
core size 172 of 198 vol = 356.994
core size 171 of 198 vol = 346.567
core size 170 of 198 vol = 337.446
core size 169 of 198 vol = 326.659
core size 168 of 198 vol = 314.95
core size 167 of 198 vol = 304.127
core size 166 of 198 vol = 294.552
core size 165 of 198 vol = 285.648
core size 164 of 198 vol = 278.884
core size 163 of 198 vol = 266.765
core size 162 of 198 vol = 258.994
core size 161 of 198 vol = 247.723
core size 160 of 198 vol = 239.84
core size 159 of 198 vol = 234.963
core size 158 of 198 vol = 230.062
```

core size 157 of 198	vol = 221.985
core size 156 of 198	vol = 215.62
core size 155 of 198	vol = 206.793
core size 154 of 198	vol = 196.984
core size 153 of 198	vol = 188.539
core size 152 of 198	vol = 182.262
core size 151 of 198	vol = 176.954
core size 150 of 198	vol = 170.712
core size 149 of 198	vol = 166.119
core size 148 of 198	vol = 159.796
core size 147 of 198	vol = 153.767
core size 146 of 198	vol = 149.092
core size 145 of 198	vol = 143.657
core size 144 of 198	vol = 137.138
core size 143 of 198	vol = 132.517
core size 142 of 198	vol = 127.231
core size 141 of 198	vol = 121.574
core size 140 of 198	vol = 116.775
core size 139 of 198	vol = 112.57
core size 138 of 198	vol = 108.17
core size 137 of 198	vol = 105.133
core size 136 of 198	vol = 101.249
core size 135 of 198	vol = 97.374
core size 134 of 198	vol = 92.974
core size 133 of 198	vol = 88.184
core size 132 of 198	vol = 84.029
core size 131 of 198	vol = 81.898
core size 130 of 198	vol = 78.019
core size 129 of 198	vol = 75.272
core size 128 of 198	vol = 73.052
core size 127 of 198	vol = 70.695
core size 126 of 198	vol = 68.975
core size 125 of 198	vol = 66.694
core size 124 of 198	vol = 64.394
core size 123 of 198	vol = 62.092
core size 122 of 198	vol = 59.045
core size 121 of 198	vol = 56.629
core size 120 of 198	vol = 54.016
core size 119 of 198	vol = 51.806
core size 118 of 198	vol = 49.652
core size 117 of 198	vol = 48.193
core size 116 of 198	vol = 46.648
core size 115 of 198	vol = 44.752

```

core size 114 of 198 vol = 43.292
core size 113 of 198 vol = 41.093
core size 112 of 198 vol = 39.147
core size 111 of 198 vol = 36.472
core size 110 of 198 vol = 34.117
core size 109 of 198 vol = 31.47
core size 108 of 198 vol = 29.448
core size 107 of 198 vol = 27.325
core size 106 of 198 vol = 25.822
core size 105 of 198 vol = 24.15
core size 104 of 198 vol = 22.648
core size 103 of 198 vol = 21.069
core size 102 of 198 vol = 19.953
core size 101 of 198 vol = 18.3
core size 100 of 198 vol = 15.723
core size 99 of 198 vol = 14.841
core size 98 of 198 vol = 11.646
core size 97 of 198 vol = 9.434
core size 96 of 198 vol = 7.354
core size 95 of 198 vol = 6.179
core size 94 of 198 vol = 5.666
core size 93 of 198 vol = 4.705
core size 92 of 198 vol = 3.665
core size 91 of 198 vol = 2.77
core size 90 of 198 vol = 2.151
core size 89 of 198 vol = 1.715
core size 88 of 198 vol = 1.15
core size 87 of 198 vol = 0.874
core size 86 of 198 vol = 0.685
core size 85 of 198 vol = 0.528
core size 84 of 198 vol = 0.37
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44

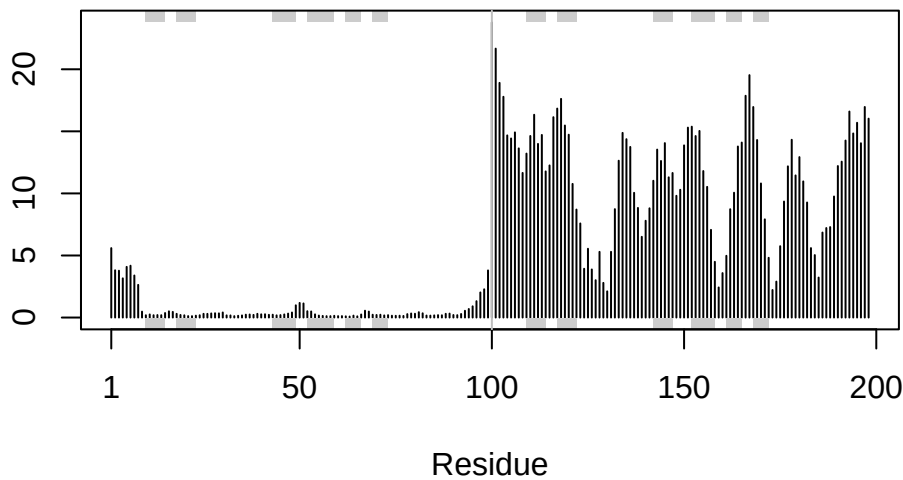
```

```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

Now we can examine the RMSF between positions of the structure. RMSF is an often used measure of conformational variance along the structure:

```
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Residue conservation from alignment file

Find the large AlphaFold alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                       full.names = TRUE)
aln_file
```

```
[1] "HIVPR_dimer_23119/HIVPR_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

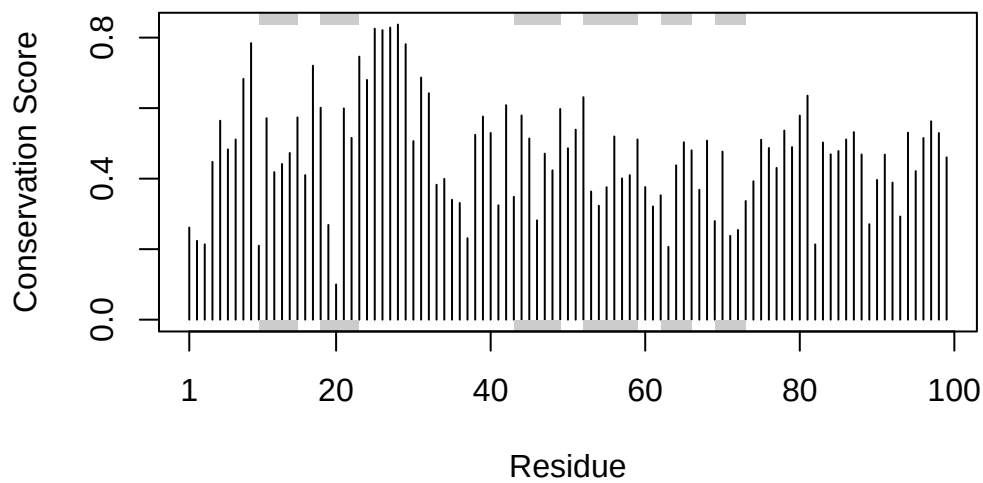
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
        ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "
```

[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"